

Shubham Sonara

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Summary

A results-driven Mechanical engineer with Proficiency in advanced Design Methodologies, system optimization, and the application of Cutting-edge engineering principles. Skilled in leveraging CAD tools, finite element analysis (FEA), and Computational fluid dynamics (CFD) to develop high-performance solutions. Specializing in sustainable Design practices, precision manufacturing, and the integration of innovative technologies to enhance operational efficiency and address Complex engineering challenges across the Automotive and Manufacturing sectors.

Education

University of New Haven

Masters in Mechanical Engineering

Expected December 2026

West Haven, CT

LJ Institute of Engineering and Technology

Bachelor of Engineering in Mechanical Engineering (GPA: 3.6/ 4.00)

July 2019 – May 2023

Ahmedabad, Gujarat

Projects

Hydraulic Double Stand for Two-Wheeler | LJJET

March 2022 – June 2022

- Designed a Cutting-edge Hydraulic Double-stand system that significantly improved parking **Efficiency** and **reduced physical effort** for two-wheeler riders, resulting in a **30% reduction in user strain**.
- Conducted advanced stress and strain analysis through simulation tools, ensuring a **100%** compliance with **Safety** and **Performance** standards under **Varying load conditions**.

Experience

Ronak Pump & Valve Pvt. Ltd.

Mechanical Design Engineer

June 2024 – Dec 2024

Ahmedabad, Gujarat

- Orchestrated the Creation of Comprehensive **engineering drawings** and Meticulously structured **Bills of Materials** to streamline the manufacturing of **Centrifugal, Diaphragm, and Vertical pumps**, Driving a remarkable **25%** reduction in production cycle time.
- Engineered and optimized high-precision pump components and assemblies leveraging advanced tools like **SolidWorks, Fusion360** and **AutoCAD**, ensuring unwavering adherence to stringent industry regulations and standards.
- Incorporated sophisticated design refinements informed by client insights and performance analyses to elevate product **Functionality, Durability, and Operational Reliability**.

Ingersoll Rand Ind Pvt. Ltd.

Apprentice Engineer at After-Market Operations Department

July 2023 – July 2024

Ahmedabad, Gujarat

- Authored and rigorously maintained exhaustive **Technical Documentation**, encompassing intricate **2D schematics**, detailed **Inventory transaction records**, **Work order logs**, and **Daily shipment data**, all within sophisticated system architectures.
- Optimized cross-functional collaboration with manufacturing and procurement teams to enhance the **Viability** and **Cost-efficiency** of aftermarket components, achieving a **10%** reduction in production delays and driving a **5%** annual cost reduction.

Ruby Bus Pvt. Ltd.

Design Engineering Intern

April 2023 – July 2023

Ahmedabad, Gujarat

- Spearheaded the comprehensive design and optimization of truck mechanical systems, Achieving a **20%** enhancement in **Performance** and **Safety standards**. Responsibilities encompassed the development of **Chassis, Bus body, and Suspension systems**, ensuring superior vehicle **Functionality** and **Seamless mobility**.

Extra-Curricular

Steering and Suspension | LJJET- Team Black hawks

Aug 2022 – May 2023

- Designed and enhanced suspension and steering systems for an off-road All-Terrain Vehicle (ATV) as part of the **SAE BAJA** competition, utilizing advanced **CAD** platforms such as **SolidWorks, MATLAB**, and **Fusion 360** for precise **3D modeling** and **simulation**. Achieved a **15%** improvement in system **Performance metrics**, including **Stability** and **Maneuverability**.
- Partnered with multidisciplinary teams to seamlessly integrate suspension and steering systems with the **Chassis** and **Drivetrain**, ensuring **optimal alignment** and **Functionality**. Contributed to achieving **100%** of project deadlines, a **20%** reduction in **Design errors**, and successful participation in **SAE BAJA** events.

Library Assistant | LJJET

Jan 2022 – July 2022

- Optimized library resource management, ensuring **Accurate cataloging** and **Enhanced accessibility** for both students and faculty. **Managed operational tasks** during academic events, **maintaining strict Adherence to library policies**, that contributed to a **30%** increase in student satisfaction and a conducive study environment.
- **Supervised** and **trained** a team of new student assistants, **elevating service standards** and achieving reduction in response time for user inquiries. Provided expert guidance in **Navigating** digital databases and physical collections, increasing student utilization of library resources by **20%**.

Certifications

- Successfully completed the hands-on course **Solid-works 3D, Solid-Edge, Auto-cad 2D** and **Fusion360** in January 2022, organized by the Department of **Mechanical Engineering** at **LJ Institute of Engineering and Technology**.